

A LOCALLY FINITE OBSTRUCTION TO LITERAL FERMIONIC COMPLETENESS

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ABSTRACT. We distinguish the unrestricted inverse-limit space of charge-zero restricted sine–Gordon towers from the algebraic graded direct sum. Under the literal componentwise definition, the former is strictly larger than the finite module generated by fermionic coefficients and odd local integrals. For odd power sums $p_{r,n}$, let $H_{k,n} = \det(p_{2(i+j)+1,n})_{0 \leq i,j < 2k+1}$. Moment-matrix factorization gives $H_{k,n} = 0$ for $k \geq n$ and $H_{k,k+1} \neq 0$. Thus $K_n = M_{0,n} \sum_{k=0}^{n-1} H_{k,n}$ is componentwise polynomial and obeys the annihilation recurrence, but contains nonzero shifted degrees $8k^2 + 6k + 1$ for all k . Every finite fermionic module expression has finite shifted-degree support, so K is outside that module. An independent Hamel-dimension argument gives the same strict containment. The obstruction does not decide completeness after imposing finite-grade support.

1. INTRODUCTION AND CATEGORY BOUNDARY

The subject rests on the exactly solved lattice models and factorized scattering theory of the nineteen-seventies and eighties [4, 32], and on the quantum inverse scattering method that exhibited their common algebra [28, 23], out of which the integrable structure of the sine–Gordon and related field theories grew. Conformal field theory supplied the second ingredient [8], with the structure of its Virasoro modules — the graded spaces a completeness statement ultimately counts — determined by Feigin and Fuchs [19], and with the massive integrable models obtained from it by perturbations that retain infinitely many integrals of motion [30]. The integrable structure common to the two was developed as a quantum Korteweg–de Vries theory in the series of Bazhanov, Lukyanov and Zamolodchikov [5, 6, 7], with the form-factor programme for massive integrable models set out by Smirnov [29] and the sine–Gordon expectation values and thermodynamics developed alongside it [24, 25, 16, 31, 18]. Null-vector and integrable-structure constructions supply the older field-theoretic setting for the tower relations [3].

Fermionic creation operators then organized correlation functions and descendant spaces across integrable lattice, conformal, and field theories, in a programme running from the XXZ chain through its scaling limits and on to conformal and sine–Gordon models [11, 12, 13, 9, 21, 22, 10, 15, 26, 14]. Jimbo, Miwa, and Smirnov formulated a restricted sine–Gordon tower space and proposed generation by finitely many fermionic coefficients over the polynomial ring of odd local integrals [20]. Subsequent work has obtained finite-level and special parameter results [1, 2].

The literal tower definition is componentwise: every particle-number component is a Laurent polynomial satisfying its degree and recurrence conditions. It does not itself impose a global finite-grade bound. This distinction changes the completeness question from an algebraic direct sum to an inverse product. The construction below is locally finite in particle number but has infinitely many global grades, which is what separates the three spaces of Table 1.

Theorem 1.1. *Fix a generic irrational $\nu \in (0, 1)$. In the literal charge-zero restricted tower space, define*

$$K_n = M_{0,n} \sum_{k=0}^{n-1} \det(p_{2(i+j)+1,n})_{0 \leq i,j < 2k+1}.$$

TABLE 1. Where the completeness question can be asked, and whether the tower K constructed below lies there. Only the first row reads the definition literally.

Space	Global grade support	Contains the tower K
Literal componentwise tower space	unrestricted	yes
Algebraic graded direct sum	finite	no
Finite fermionic module	finite	no

Then K is an admitted nonzero tower but is not a finite combination, over the polynomial ring generated by the odd local integrals, of fixed joint Laurent coefficients of balanced fermionic creation-operator products. This holds for either standard Laurent expansion chosen uniformly at zero or at infinity.

2. TOWER OBJECTS AND THE MISSING GLOBAL RESTRICTION

Fix a generic irrational $\nu \in (0, 1)$ throughout.

Definition 2.1. For $l - n = c$, a *charge- c tower* is a family of Laurent-polynomial components

$$L^{(l,n)}(S_1, \dots, S_l \mid B_1, \dots, B_{2n}),$$

antisymmetric in the S variables and symmetric in the B variables, subject to the componentwise degree condition $0 \leq \deg_{S_i} L^{(l,n)} \leq 2n - 1$ and the annihilation recurrence

$$\begin{aligned} L^{(l,n)}(S_1, \dots, S_{l-1}, B \mid B_1, \dots, B_{2n-2}, B, -B) \\ = (-1)^c B \prod_{a=1}^{l-1} (B^2 - S_a^2) L^{(l-1,n-1)}(S_1, \dots, S_{l-1} \mid B_1, \dots, B_{2n-2}). \end{aligned} \quad (2.1)$$

Write $\mathcal{T}_{0,c}$ for the space of charge- c towers.

The primary charge-zero tower is

$$M_{0,n}(S_1, \dots, S_n) = \det(S_a^{2b-1})_{1 \leq a, b \leq n}, \quad M_{0,0} = 1, \quad (2.2)$$

in which the vacuum expectation value of the primary field, an overall constant carried explicitly in [20] and evaluated in [25], is set to one; it multiplies every component alike and no statement below depends on it. The odd local integrals act on components by multiplication by

$$I_{2j-1,n} = \sum_{a=1}^{2n} B_a^{2j-1}, \quad \bar{I}_{2j-1,n} = \sum_{a=1}^{2n} B_a^{-(2j-1)}. \quad (2.3)$$

These are the odd local integrals of motion of the model, whose construction for quantum sine-Gordon and for perturbed conformal field theory goes back to [27, 17, 30]; on a tower each acts separately on every component, so the polynomial ring they generate acts by multiplication and moves the joint degree by odd amounts.

With $P_n(S) = \prod_a (S - B_a)$, the creation operators are

$$\begin{aligned} (\psi_0^*(Z)L)^{(l+1,n)} &= \frac{\text{Skew}_{S_0, \dots, S_l}(C_n(Z, S_0)L^{(l,n)}(S_1, \dots, S_l))}{P_n(-Z)l!}, \\ (\chi_0^*(Z)L)^{(l-1,n)} &= \frac{L^{(l,n)}(Z, S_1, \dots) - L^{(l,n)}(-Z, S_1, \dots)}{2P_n(-Z)}, \end{aligned} \quad (2.4)$$

where Skew is unnormalized antisymmetrization in the displayed S variables,

$$\text{Skew}_{S_0, \dots, S_l} f = \sum_{\sigma \in \mathfrak{S}_{l+1}} \text{sgn}(\sigma) f(S_{\sigma(0)}, \dots, S_{\sigma(l)}),$$

and the kernel is the signed expression

$$C_n(S_1, S_2) = \frac{S_1}{4\nu} \sum_{\epsilon_1, \epsilon_2 = \pm 1} \frac{P_n(\epsilon_1 S_1) P_n(\epsilon_2 S_2)}{\epsilon_1 S_1 + \epsilon_2 S_2}, \quad (2.5)$$

a polynomial of joint degree $4n$.

The statement at issue asserts that every $L \in \mathcal{T}_{0,0}$ is a *finite* combination, over the polynomial ring generated by (2.3), of fixed joint formal Laurent coefficients of balanced products of the operators (2.4) acting on M_0 .

Two features of that formulation are what the counterexample uses, and both are visible in the source. First, $\mathcal{T}_{0,c}$ is defined componentwise in [20]: each particle-number component is separately a Laurent polynomial of the stated degree obeying (2.1), and neither the definition nor the completeness proposal imposes a uniform total B -degree or a finite-grade support across n . Second, the denominator in (2.4) leaves the expansion point under-specified; the argument below works for either standard Laurent expansion, at zero or at infinity, chosen uniformly for the tower-valued series. A mode exponent that depends on n is not a coefficient of one formal Laurent series, so that reading is not available.

3. ODD-MOMENT HANKEL DETERMINANTS

For positive odd r write

$$p_{r,n} = \sum_{a=1}^{2n} B_a^r = I_{r,n}, \quad (3.1)$$

and for $k \geq 0$ set $m = 2k + 1$ and

$$H_{k,n} = \det(p_{2(i+j)+1,n})_{0 \leq i,j < m}. \quad (3.2)$$

Lemma 3.1. *Every $H_{k,n}$ is stable under insertion of the annihilation pair:*

$$p_{r,n}(B_1, \dots, B_{2n-2}, B, -B) = p_{r,n-1}(B_1, \dots, B_{2n-2}) \quad (3.3)$$

for odd r , hence $H_{k,n}(B_1, \dots, B_{2n-2}, B, -B) = H_{k,n-1}$.

Proof. For odd r the two inserted variables contribute $B^r + (-B)^r = 0$. Substituting into every entry of (3.2) gives the second assertion. $\square$

Lemma 3.2. *$H_{k,n} = 0$ whenever $k \geq n$, and $H_{k,k+1} \neq 0$.*

Proof. Let V be the $m \times 2n$ matrix $V_{i,a} = B_a^{2i}$ and $D = \text{diag}(B_1, \dots, B_{2n})$. The matrix in (3.2) is $VDV^\top$. If $k \geq n$ then $m = 2k + 1 > 2n$, so the rank of $VDV^\top$ is at most $2n < m$ and the determinant vanishes.

The threshold is sharp. At $n = k + 1$ there are $m + 1$ particle variables; set one of them to zero and choose the remaining m with nonzero distinct squares. Then

$$H_{k,k+1} = \left(\prod_{a=1}^m B_a \right) \left(\prod_{1 \leq a < b \leq m} (B_b^2 - B_a^2) \right)^2 \neq 0, \quad (3.4)$$

so the polynomial $H_{k,k+1}$ is not identically zero. Alternatively, choosing all $m + 1$ variables distinct and positive makes $VDV^\top$ positive definite, which exhibits a nonzero value inside $(\mathbb{C}^*)^{2k+2}$. $\square$

Lemma 3.3. *$H_{k,n}$ is homogeneous of degree*

$$d_k = 2m^2 - m = 8k^2 + 6k + 1, \quad (3.5)$$

and $(d_k)_{k \geq 0}$ is strictly increasing.

Proof. Every monomial of the determinant (3.2) is a product of m entries $p_{2(i+\sigma(i))+1,n}$ over a permutation σ of $\{0, \dots, m-1\}$, so its degree is $\sum_{i=0}^{m-1} (2(i+\sigma(i))+1)$. Since $\sum_i (i+\sigma(i)) = m(m-1)$ independently of σ , every monomial carries the same degree $2m(m-1) + m$, which is (3.5). Strict monotonicity is immediate from $d_{k+1} - d_k = 16k + 14 > 0$. $\square$

4. AN ADMITTED LOCALLY FINITE TOWER

Definition 4.1. Put

$$F_0 = 0, \quad F_n = \sum_{k=0}^{n-1} H_{k,n}, \quad K_n = M_{0,n} F_n. \quad (4.1)$$

By Lemma 3.2, (4.1) is also the componentwise value of the formal sum $M_{0,n} \sum_{k \geq 0} H_{k,n}$, every component of which is an ordinary finite polynomial. This is the sense in which K is locally finite and globally unbounded, and it is the whole construction.

Proposition 4.2. $K \in \mathcal{T}_{0,0}$, and $K \neq 0$.

Proof. The primary determinant factorizes as

$$M_{0,n}(S_1, \dots, S_{n-1}, B) = B \prod_{a=1}^{n-1} (B^2 - S_a^2) M_{0,n-1}(S_1, \dots, S_{n-1}), \quad (4.2)$$

is alternating in S , and has $\deg_{S_i} \leq 2n-1$. The multiplier F_n is symmetric in B and free of S , so K_n has all the symmetries and degrees required by Definition 2.1.

Under insertion of the annihilation pair, Lemmas 3.1 and 3.2 give

$$F_n(B_1, \dots, B_{2n-2}, B, -B) = \sum_{k=0}^{n-1} H_{k,n-1} = \sum_{k=0}^{n-2} H_{k,n-1} = F_{n-1}, \quad (4.3)$$

the removed term being $H_{n-1,n-1} = 0$. Combining (4.2) and (4.3) gives (2.1) with $c = 0$ for every n . Finally $K_1(S_1 \mid B_1, B_2) = S_1(B_1 + B_2) \neq 0$. $\square$

Corollary 4.3. For every k the component K_{k+1} carries a nonzero homogeneous term $M_{0,k+1} H_{k,k+1}$, whose degree relative to the primary is d_k . Hence K carries infinitely many distinct nonzero shifted degrees.

Proof. Immediate from (3.4) and Lemma 3.3, the degrees d_k being distinct by strict monotonicity. $\square$

5. FINITE FERMIONIC EXPRESSIONS HAVE FINITE SUPPORT

Lemma 5.1. Define the degree baseline

$$b(l, n) = 2nl - n^2. \quad (5.1)$$

Then the primary (2.2) has degree $b(n, n) = n^2$, and the operators (2.4) shift the joint degree by exactly the baseline increments

$$b(l+1, n) - b(l, n) = 2n, \quad b(l-1, n) - b(l, n) = -2n. \quad (5.2)$$

Proof. The kernel C_n has joint degree $4n$ by (2.5) and $P_n(-Z)$ has degree $2n$, so $\psi_0^*(Z)$ raises the joint degree by $2n$; the numerator of $\chi_0^*(Z)$ leaves the degree unchanged while the same denominator lowers it by $2n$. These agree with (5.2). $\square$

Proposition 5.2. Every finite combination, over the polynomial ring generated by (2.3), of fixed joint Laurent coefficients of balanced products of (2.4) acting on M_0 has finitely many shifted degrees.

Proof. Let $G(Z; S, B) = \sum_a Z^a G_a(S, B)$ be jointly homogeneous. Comparing the two sides after simultaneous scaling of all variables shows that each fixed coefficient G_a is homogeneous, of the joint degree minus a . The same comparison applies to any finite list of variables Z_i, X_i and to the expansion at zero or at infinity alike. With [Lemma 5.1](#), every fixed joint coefficient of a balanced fermionic product acting on M_0 therefore carries a single degree relative to (5.1), independent of the particle number.

Each I_{2j-1} shifts that degree by $2j-1$ and each $\bar{I}_{2j-1}$ by $-(2j-1)$, and a polynomial has finitely many monomials. So a finite combination occupies finitely many shifted degrees. $\square$

6. THE CONTRADICTION, AND AN INDEPENDENT DIMENSION CHECK

Proof of Theorem 1.1. [Proposition 4.2](#) places K in $\mathcal{T}_{0,0}$ and shows it is nonzero. [Corollary 4.3](#) gives it infinitely many distinct nonzero shifted degrees, while [Proposition 5.2](#) allows every element of the proposed module only finitely many. Hence K is not such a combination, for either uniform choice of expansion point, and for every admissible generic irrational ν . $\square$

The conclusion does not depend on the particular tower K , nor on the choice of expansion point, as the following count shows independently.

Proposition 6.1. *$\mathcal{T}_{0,0}$ has uncountable Hamel dimension, whereas the proposed module is countably spanned. The containment is therefore strict.*

Proof. For a sequence $a = (a_0, a_1, \dots) \in \mathbb{C}^{\mathbb{N}}$ put

$$K_n^a = M_{0,n} \sum_{k=0}^{n-1} a_k H_{k,n}, \quad (6.1)$$

which is an admitted tower by the proof of [Proposition 4.2](#). If k is the first index with $a_k \neq 0$, component $k+1$ carries the nonzero degree- d_k term $a_k M_{0,k+1} H_{k,k+1}$, so $a \mapsto K^a$ is injective. The sequences $(1, \lambda, \lambda^2, \dots)$ for distinct $\lambda \in \mathbb{C}$ are linearly independent by the finite Vandermonde determinant, and there are uncountably many of them.

The proposed module, by contrast, is spanned by a countable set: the balancing index r is an integer, the fixed coefficient multi-indices lie in $\mathbb{Z}^{2r}$, and the monomials in the countable family of integral generators form a countable set. $\square$

7. THE REPAIRED CONJECTURE, WHICH THIS PAPER LEAVES OPEN

The natural repair replaces the unrestricted componentwise tower space by its algebraic graded direct sum: require finite shifted-degree support, or formulate completeness one homogeneous grade at a time. Either condition excludes (4.1) and (6.1), and either matches the reading suggested by finite-level descendant spaces.

This paper neither proves nor disproves that repaired statement. The completeness task is posed in [20]; with the local integrals of motion acting nontrivially, the identification with a Virasoro basis has been carried out only to level four [14], later work on the underlying expectation values does not raise that level [1], and the full construction of [2] is confined to the free-fermion point. The intended finite-grade conjecture therefore remains open.

8. CONCLUSION

Literal inverse-limit completeness fails because componentwise finiteness does not imply finite global grade. Requiring finite shifted-degree support removes the constructed obstruction and matches the usual algebraic descendant interpretation. Whether the fermionic generators are complete in that repaired finite-grade space remains open.

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